Midterm Review: Scatterplots and Linear Models

MATH 213

Group 1

I have a dataset that contains information about all 58 counties in California. The data were gathered between 2006-2010. Here is a part of the code book for this dataset.

name: County name

metro: Whether the county contains a metropolitan area, with levels NO, YES

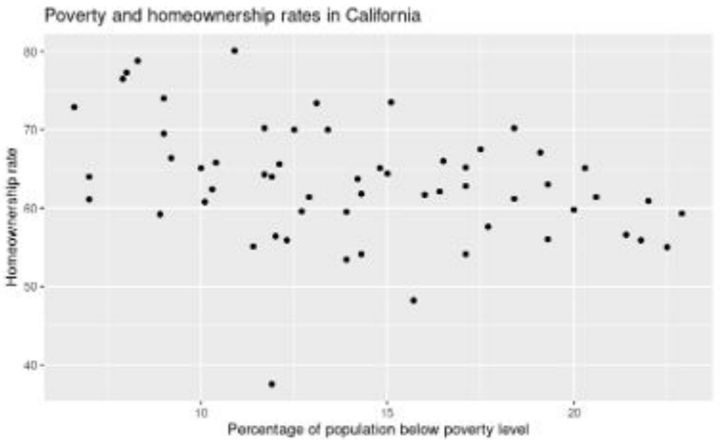
poverty: Percent of population in the county below poverty level (2006-2010)

homeownership: Home ownership rate for the county (2006 – 2010)

and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
Amador County	NO	8.0	77.3

Here is a graph of “homeownership” versus “poverty”



1. What are the cases in this study?

The county’s in California.

2. Write a paragraph describing the scatterplot in context. Start with a WWWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

The scatterplot represents the 58 counties in California with the people's home ownership status being compared to the percentage of people living under the poverty line while also being in a metro county or not. There is a moderate negative relationship for homeownership going down as the poverty level increases.

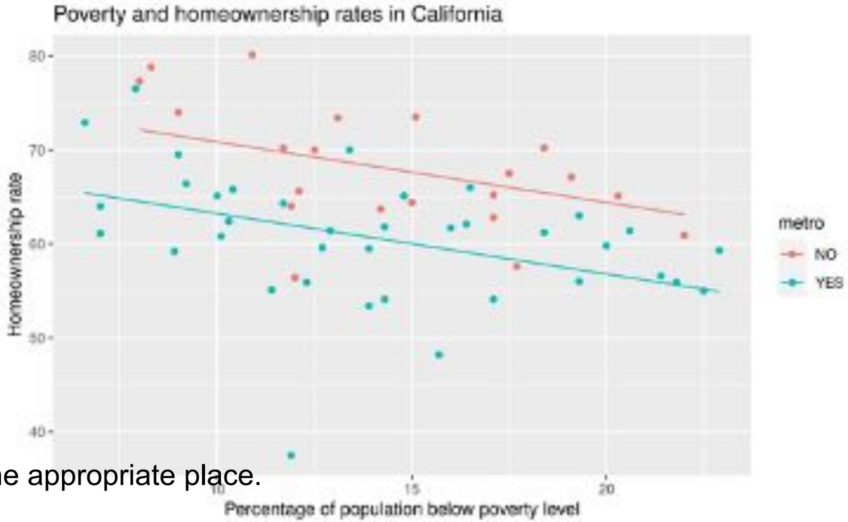
I had R create a model for predicting “homeownership” using “poverty” and “metro” as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_complete2

Residuals:
     Min       1Q   Median       3Q      Max
-24.5043  -3.4089   0.7394   4.1270  11.9210

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  77.2778     2.9983  25.774  < 2e-16 ***
poverty      -0.6437     0.1855  -3.469  0.00102 **
metroYES     -7.6137     1.6913  -4.502 3.54e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.186 on 55 degrees of freedom
Multiple R-squared:  0.3621,    Adjusted R-squared:  0.3389
F-statistic: 15.61 on 2 and 55 DF,  p-value: 4.268e-06
```



3. Write the formula for the regression equation with a “hat” in the appropriate place.

p^ = 77.2778 + -0.6437(poverty) + -7.6137(metroYES)

4. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

The model predicts that LA will have a homeownership rate of 59.6632

5. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county? Does that mean LA county's data point is above or below the line? Does that mean LA County has more homeowners than predicted or fewer?

The residual for this county is P - p^ => 48.2% - 59.62% = -11.42

. This means that LA county's datapoint is above the line. This means that LA Count has more homeowners than predicted.

6. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write, and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1:

Intercept: The predicted based homeownership rate of a non-metropolitan county in california is 77.2778 if there is a poverty rate of 0%

Sentence 2:

poverty: For every increase in percent of population below poverty line, the predicted percentage of homeownership decreases by 0.6437, metro or not

Sentence 3:

metroYES: If the county has a metropolitan area, the predicted percentage of homeownership decreases by 7.6137

7. Interpret the R^2 value for this model.

Group 2

Here is a graph of “homeownership” versus “poverty”

I have a dataset that contains information about all 58 counties in California. The data were gathered between 2006-2010. Here is a part of the code book for this dataset.

name: County name

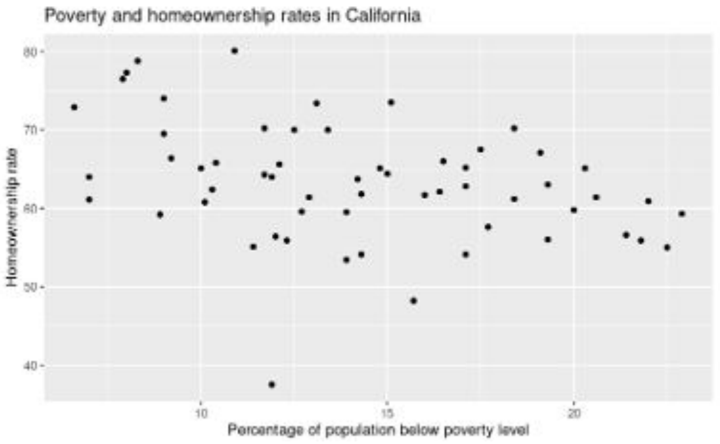
metro: Whether the county contains a metropolitan area, with levels NO, YES

poverty: Percent of population in the county below poverty level (2006-2010)

homeownership: Home ownership rate for the county (2006 – 2010)

and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
Amador County	NO	8.0	77.3



1. What are the cases in this study?

The 58 counties in California.

2. Write a paragraph describing the scatterplot in context. Start with a WWWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

The scatterplot shows the relationship between homeownership rate and the percentage of population below the poverty level of 58 counties in California. There is a moderate negative linear association between these two variables, which means that as a greater percentage of the population is under the poverty level, the homeownership rate declines.

For example, when the percentage of the population under the poverty level is about 7%, the homeownership rate is as high as 80%, but decreases to about 55% when the percentage of the population under the poverty level is 25%.

However, there are counties where the poverty percentage is still about 7% but the homeownership rate is as low as 60%.

I had R create a model for predicting “homeownership” using “poverty” and “metro” as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_complete2)

Residuals:
     Min       1Q   Median       3Q      Max
-24.5043  -3.4089   0.7394   4.1270  11.9210

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  77.2778     2.9983  25.774  < 2e-16 ***
poverty      -0.6437     0.1855  -3.469  0.00102 **
metroYES     -7.6137     1.6913  -4.502 3.54e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.186 on 55 degrees of freedom
Multiple R-squared:  0.3621,	Adjusted R-squared:  0.3389
F-statistic: 15.61 on 2 and 55 DF,  p-value: 4.268e-06
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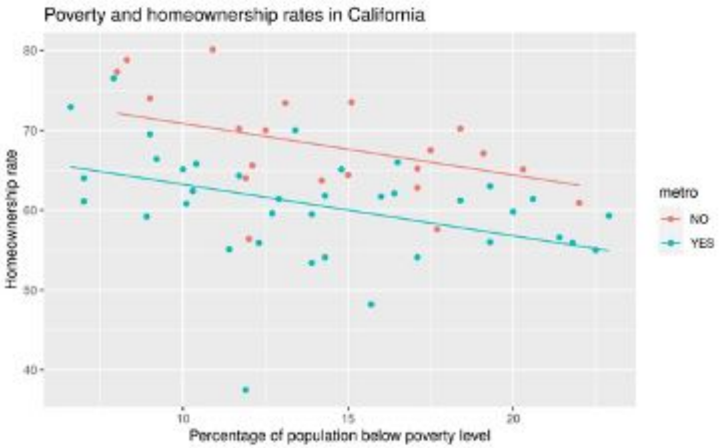


3. Write the formula for the regression equation with a “hat” in the appropriate place.

y_hat = 77.2778 + (-0.6437x_1 - 7.6137x_2)

4. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

59.56 % ?

5. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county? Does that mean LA county's data point is above or below the line? Does that mean LA County has more homeowners than predicted or fewer?

Residuals calculate the actual value minus the predicted value. 48.2 - 59.56 = -11.36

This means that LA county is below the line; LA county has less homeowners than the predicted value.

6. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write, and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1: (Intercept) When there is a 0% poverty rate for a county the homeownership rate starts as high as 77.27%

Sentence 2: (Poverty) When the poverty rate increases by 1%, the predicted homeownership rate decreases by 0.64%.

Sentence 3: (MetroYES) If the county has a metropolitan area, the predicted homeownership rate decreases by 7.61%.

7. Interpret the R^2 value for this model.

The R^2 value for this model is 0.3389. This means that about 34% of the variance in data is explained by the model.

Group 3

I have a dataset that contains information about all 58 counties in California. The data were gathered between 2006-2010. Here is a part of the code book for this dataset.

name: County name

metro: Whether the county contains a metropolitan area, with levels NO, YES

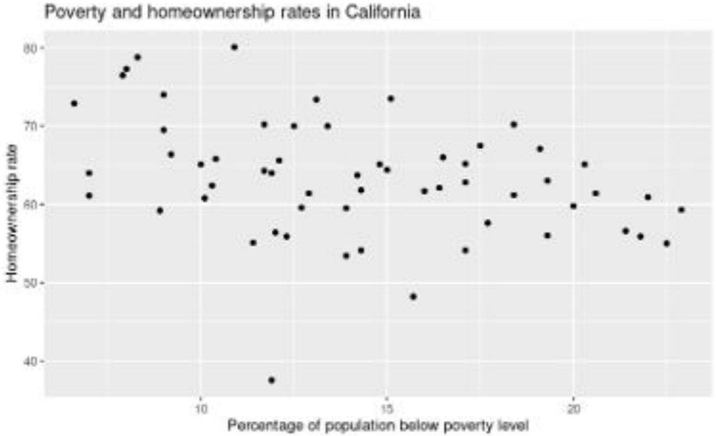
poverty: Percent of population in the county below poverty level (2006-2010)

homeownership: Home ownership rate for the county (2006 – 2010)

and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
Amador County	NO	8.0	77.3

Here is a graph of “homeownership” versus “poverty”



1. What are the cases in this study?
All 58 counties in Ca.

2. Write a paragraph describing the scatterplot in context. Start with a WWWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

The graph displays data from all 58 counties in California, showing the relationship between poverty rates and homeownership rates from 2006 to 2010. There is a moderately strong, negative linear association, as the percentage of the population below the poverty level increases, the homeownership rate tends to decrease. For a county that is 10% below the poverty line, they have a 60% - 80% homeownership rate, while a county that is 25% below the poverty line, they have a 55% to 65% homeownership rate. One exception is a county with a 14% poverty rate but a much lower than expected homeownership rate of approximately 35%. Overall, the trend suggests that higher poverty is generally associated with lower homeownership in California counties during this time.

I had R create a model for predicting “homeownership” using “poverty” and “metro” as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_complete2)

Residuals:
     Min       1Q   Median       3Q      Max 
-24.5043  -3.4089   0.7394   4.1270  11.9210 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  77.2778     2.9983  25.774  < 2e-16 ***
poverty      -0.6437     0.1855  -3.469  0.00102 ** 
metroYES     -7.6137     1.6913  -4.502 3.54e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.186 on 55 degrees of freedom
Multiple R-squared:  0.3621,	Adjusted R-squared:  0.3389 
F-statistic: 15.61 on 2 and 55 DF,  p-value: 4.268e-06
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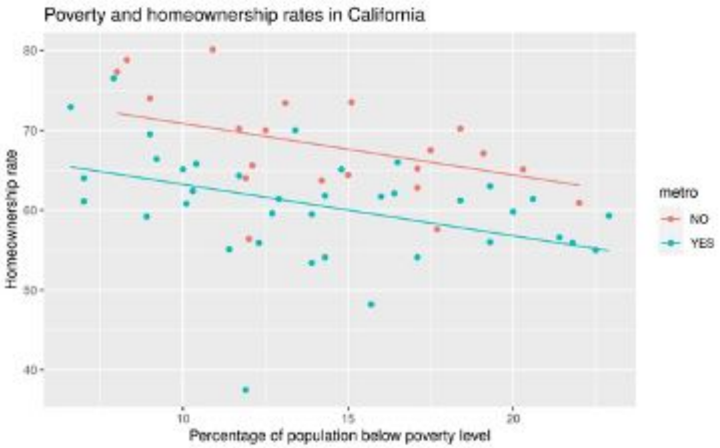


3. Write the formula for the regression equation with a “hat” in the appropriate place.

Homeownershiphat = 77.2778 - (0.6437 * poverty) - (7.6137 * metro)

4. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

Homeownershiphat = 77.2778 - (0.6437 * 15.7) - (7.6137 * 1) = 59.56

5. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county? Does that mean LA county's data point is above or below the line? Does that mean LA County has more homeowners than predicted or fewer?

The residual for this county is -11.4, which means the homeownership rate is 11.4% lower than expected. Since the value is negative the, data point for LA county is below the regression line. SO LA county has fewer predicted homeowners than the model predicted.

6. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write, and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1: The intercept is 77.28, so a county with a poverty rate of 0%, and is not metro, the predicted homeownership rate is 77.28%.

Sentence 2: The poverty coefficient of -0.6437, which means than for every 1 percent increase in poverty rates, the predicted homeownership rate decrease by 0.6437%.

Sentence 3: The metro coefficient of -7.61 means that if a county is metro, than the expected homeownership rate decreases by 7.61%.

7. Interpret the R^2 value for this model.

About 36.2% of the variation in homeownership rates in CA, counties is explained by the model that includes poverty rate and metro status as predictors. So there is about 63.8% of variation that is not accounted for by these two variables.

Group 4

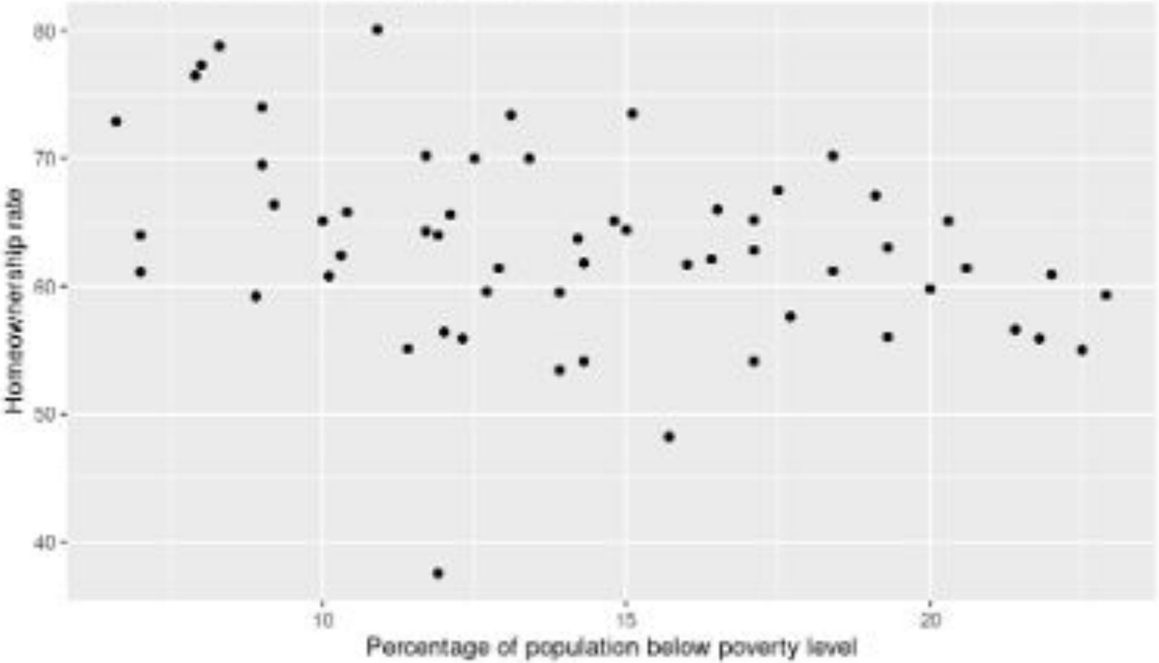
I have a dataset that contains information about
California. The data were gathered between 200
of the code book for this dataset.

name: County name

metro: Whether the county contains a me
levels NO, YES

poverty: Percent of population in the cou
(2006-2010)

homeownership: Home ownership rate f
2010)



and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
Amador County	NO	8.0	77.3

1. What are the cases in this study?

The 58 counties in California

2. Write a paragraph describing the scatterplot in context. Start with a WWWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

The scatterplot describes the homeownership rate (%) and poverty levels (%) for 58 counties in California between 2006 and 2010. The moderate correlation between the percentage of homeownership and percentage of population below the poverty line is a negative linear. The general trend has counties largely between 5% and 25% of the population below poverty level where those near the 5% mark have homeownership in the 70% to 80% homeownership and those counties with the higher poverty levels drops to only 65% to 55% of the county population owning their homes.We have a ~~For example, one county has 7% of the population below poverty level and 77% homeownership rate, while another county has 25% of the population below poverty level and 55% homeownership rate.~~ There are some exceptions to the trend, like one county with 12% of the population below poverty level and only 37% homeownership rate.

I had R create a model for predicting "homeownership" using "poverty" and "metro" as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_complete2)

Residuals:
     Min       1Q   Median       3Q      Max
-24.5043  -3.4089   0.7394   4.1270  11.9210

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  77.2778     2.9983  25.774  < 2e-16 ***
poverty      -0.6437     0.1855  -3.469  0.00102 **
metroYES     -7.6137     1.6913  -4.502 3.54e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.186 on 55 degrees of freedom
Multiple R-squared:  0.3621,    Adjusted R-squared:  0.3389
F-statistic: 15.61 on 2 and 55 DF,  p-value: 4.268e-06
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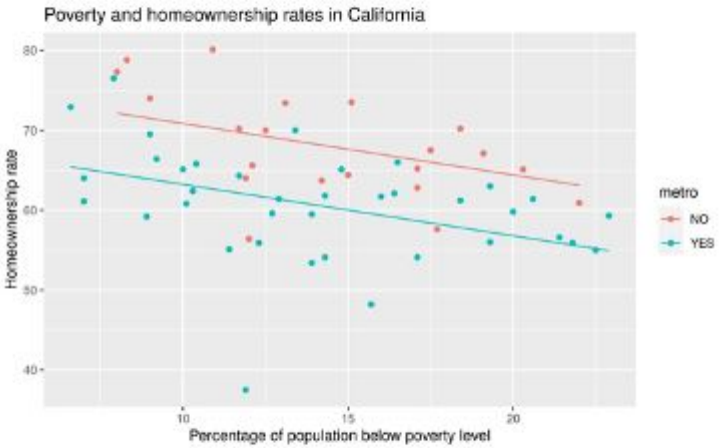


3. Write the formula for the regression equation with a "hat" in the appropriate place.

$\hat{P}$ = 77.28 - 0.64(poverty) - 7.61(metroYes)

4. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

$\hat{P}$ = 77.28 - .64(15.7) - 7.61 = 59.62%

5. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county? Does that mean LA county's data point is above or below the line? Does that mean LA County has more homeowners than predicted or fewer?

Residual is P - $\hat{p}$ => 48.2% - 59.62% = -11.42

6. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write, and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1:

Intercept - This is the intercept at which there is no poverty and no metro area in the county and states that there should be roughly 77.28% of the population that owns their home.

Sentence 2:

Poverty - This is the slope adjustment for poverty levels in the Ca counties. It shows that for every percent increase in poverty, homeownership rates drop by .64%.

Sentence 3:

MetroYes - The predicted homeownership rate decreases by 7.61 if the county has a metro.

7. Interpret the R^2 value for this model.

About 36.21% of the variation in homeownership rate is explained by whether the county has a metro and the percentage of the county's population below poverty level.

Group 5

Here is a graph of “homeownership” versus “poverty”

I have a dataset that contains information about all 58 counties in California. The data were gathered between 2006-2010. Here is a part of the code book for this dataset.

name: County name

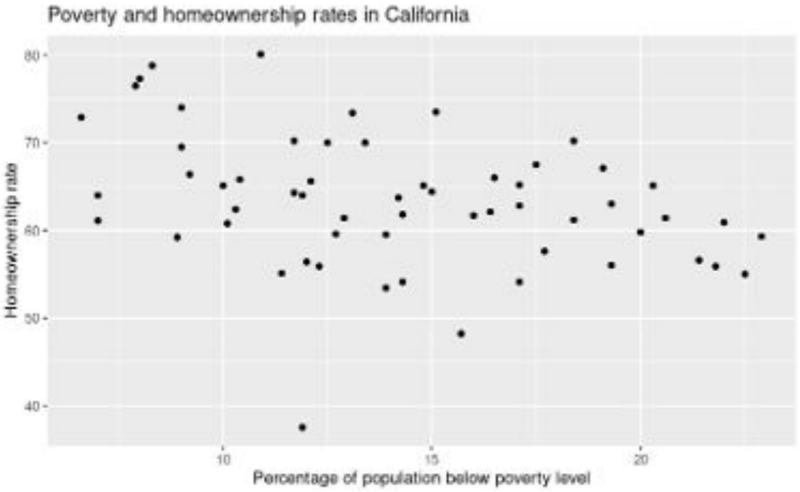
metro: Whether the county contains a metropolitan area, with levels NO, YES

poverty: Percent of population in the county below poverty level (2006-2010)

homeownership: Home ownership rate for the county (2006 – 2010)

and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
Amador County	NO	8.0	77.3



1. What are the cases in this study?

The 58 counties in California

2. Write a paragraph describing the scatterplot in context. Start with a WWWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

The scatterplot shows the percentage below poverty versus the homeownership level of all the counties in california. The shape of the graph is moderately strong with only a few outliers for example this county was 12% below poverty and had a below 40 homeownership rate. The graph has a negative linear shape with most of the counties having a homeownership rate just above 60%, with all of the counties below poverty percentage being between 5% to 20%. The graph gets its negative trend because as the below poverty level drops so does the homeownership.

I had R create a model for predicting “homeownership” using “poverty” and “metro” as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_comp

Residuals:
     Min       1Q   Median       3Q      Max
-24.5043  -3.4089   0.7394   4.1270  11.9210

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  77.2778     2.9983  25.774  < 2e-16 ***
poverty      -0.6437     0.1855  -3.469  0.00102 **
metroYES     -7.6137     1.6913  -4.502 3.54e-05 ***
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.186 on 55 degrees of freedom
Multiple R-squared:  0.3621,	Adjusted R-squared:  0.3389
F-statistic: 15.61 on 2 and 55 DF,  p-value: 4.268e-06
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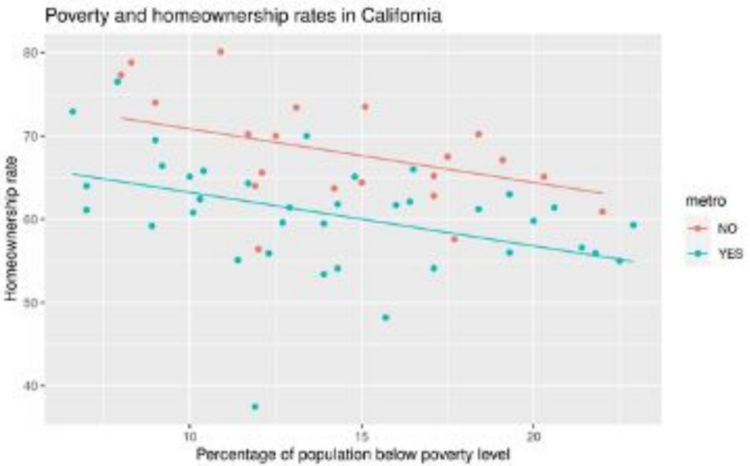

3. Write the formula for the regression equation with a “hat” in the appropriate place.

homeownership-hat=77.2778+-0.6437(poverty percentage)+-7.6137(yes=1 no=0)

4. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

69.0204

5. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county? Does that mean LA county's data point is above or below the line? Does that mean LA County has more homeowners than predicted or fewer?

The data point is above the line meaning that the predicted value has more homeowners than the actual LA count

6. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write, and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1:
A non-metro county with a 0% poverty rate has a predicted home ownership rate of 77.3%

Sentence 2:

Within either category of counties (metro or not) , a 1% increase in poverty level is associated with a 0.64% decrease in home ownership rates

Sentence 3:

Counties in a metro area have home ownership levels that are 7.6% lower than non metro counties with an equivalent poverty rate

7. Interpret the R^2 value for this model.

About 33.89% of the variability in poverty percentage can explain the homeownership rate in california.

Group 6

I have a dataset that contains information about all 58 counties in California. The data were gathered between 2006-2010. Here is a part of the code book for this dataset.

name: County name

metro: Whether the county contains a metropolitan area, with levels NO, YES

poverty: Percent of population in the county below poverty level (2006-2010)

homeownership: Home ownership rate for the county (2006 – 2010)

and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
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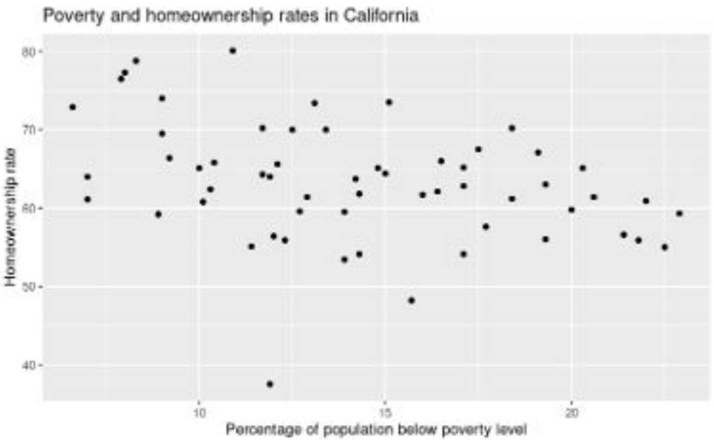
Here is a graph of “homeownership” versus “poverty”



1. What are the cases in this study?
The cases in this study are the 58 counties in California

2. Write a paragraph describing the scatterplot in context. Start with a WWWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

The information above shows 58 counties that are in California and shows the poverty level of the 58 counties in 2006-2010. The data is not concentrated anywhere on the graph but it is linear, there seems to be a negative trend between the variables of homeownership and poverty percentage. Counties with 10% of below the poverty level have a homeownership of 60% to 80%, where the counties with 20% below the poverty level have a homeownership of 55% to 65%. As the poverty percentage rises he homeownership falls as the graph goes on. All the data points are between 5% and 22% below the poverty line.

I had R create a model for predicting “homeownership” using “poverty” and “metro” as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_complete2)

Residuals:
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Coefficients:
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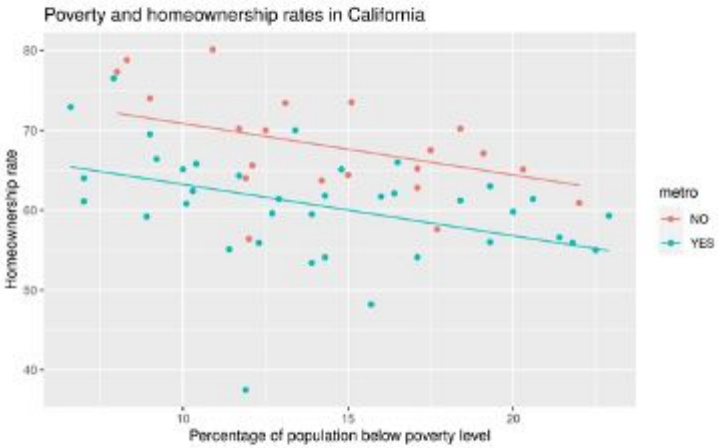


3. Write the formula for the regression equation with a “hat” in the appropriate place.

p-hat=77.277+ -7.6137(metro)+-0.64(percentage of poverty)

4. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

69.56

5. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county? Does that mean LA county's data point is above or below the line? Does that mean LA County has more homeowners than predicted or fewer?

The residual for the data is 21.36. It is below the line. They have fewer than what was predicted that they would have.

6. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write, and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1: The average homeownership when there is 0% below the poverty line is 77.277 without a metro

Sentence 2: The -7.137 shows the that if a county is in a metro area they have this level of homeownership, lower than non-metro counties with an equivalent poverty rate

Sentence 3: With either category of counties with or without a metro, for the -0.6437 shows that for every percent of poverty rate the homeownership drops by this value

7. Interpret the R^2 value for this model.

About 36% of the variability in the homeownership is explained by using the counties population below the poverty level as a predictor

Group 7

Here is a graph of “homeownership” versus “poverty”

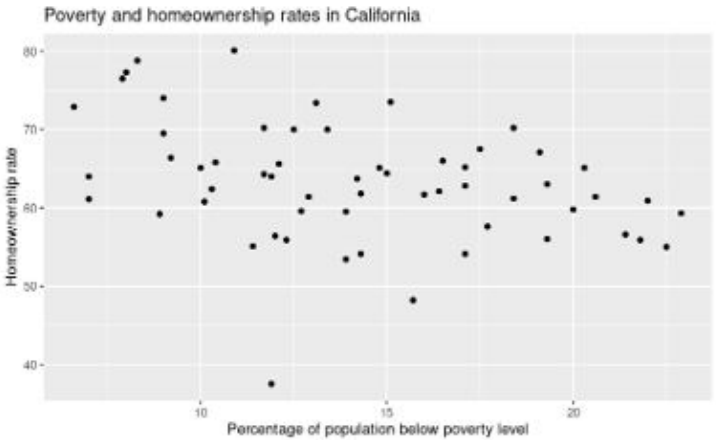
I have a dataset that contains information about all 58 counties in California. The data were gathered between 2006-2010. Here is a part of the code book for this dataset.

name: County name

metro: Whether the county contains a metropolitan area, with levels NO, YES

poverty: Percent of population in the county below poverty level (2006-2010)

homeownership: Home ownership rate for the county (2006 – 2010)



and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
Amador County	NO	8.0	77.3

1. What are the cases in this study?
   The 58 counties in california

2. Write a paragraph describing the scatterplot in context. Start with a WWWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

This scatter plot shows the relationship between the percentage of the population below the poverty level and home ownership rate. Generally there is a moderate negative linear association between these variables. For example, a country with around 8% of the population below the poverty line will have a homeownership rate of between 60 - 75%, while a country with 23% of the population below the poverty line will have a homeownership rate around 55 - 60%. Though there are exceptions, such as a country with 12% of the population below the poverty line, but only a 37% homeownership rate.

Since the model is too spread out, include a range of values for the y value

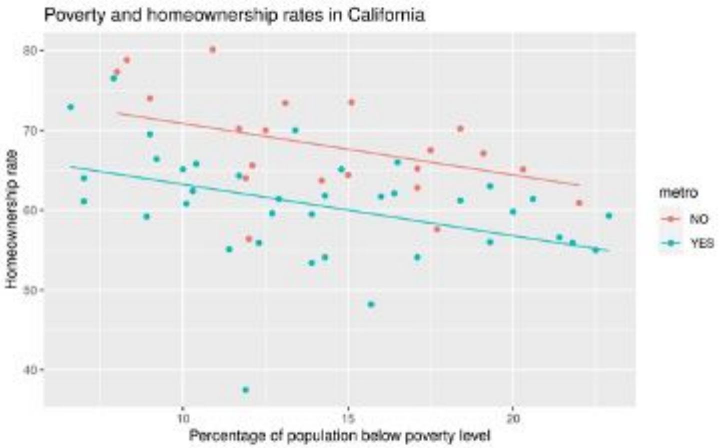
I had R create a model for predicting “homeownership” using “poverty” and “metro” as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_complete2)

Residuals:
     Min       1Q   Median       3Q      Max 
-24.5043  -3.4089   0.7394   4.1270  11.9210 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  77.2778     2.9983  25.774  < 2e-16 ***
poverty      -0.6437     0.1855  -3.469  0.00102 ** 
metroYES     -7.6137     1.6913  -4.502 3.54e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.186 on 55 degrees of freedom
Multiple R-squared:  0.3621,    Adjusted R-squared:  0.3389 
F-statistic: 15.61 on 2 and 55 DF,  p-value: 4.268e-06
```



3. Write the formula for the regression equation with a “hat” in the appropriate place.

homeowner_hat = 77.2778 - 0.6437(poverty) - 7.6137(metroYES)

4. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

homeowner_hat = 77.2778 - 0.6437(15.7) - 7.6137(1)

homeowner_hat = 59.6%

5. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county? Does that mean LA county's data point is above or below the line? Does that mean LA County has more homeowners than predicted or fewer?

Y- y_hat

48.2 - 59.6 = -11.4

LA County has less homeowners than predicted (below the regression line)

6. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write, and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1: intercept
The intercept

When the poverty rate is 0%, and the county doesn't have a metro, the predicted homeownership rate is 77.2778

Sentence 2: poverty
When the poverty rate goes up by 1%, and the country doesn't have a metro, the homeownership rate decreases by 0.6437%

Sentence 3: metroYES

When the country has a metro, and the poverty rate is 0%, the homeownership rate decreases by 7.6137%

7. Interpret the R^2 value for this model.

The model has a r^2 value of 0.3621 which means 36.21 percent of the variability in homeownership can be explained by the model(ex. By type of county and poverty rate). This suggests percentage of population below poverty level is not a great predictor of homeownership rates. However since this is not 100% or close it is likely other factors also contribute to the percentage of home ownership.

Group 8

Here is a graph of "homeownership" versus "poverty"

I have a dataset that contains information about all 58 counties in California. The data were gathered between 2006-2010. Here is a part of the code book for this dataset.

name: County name

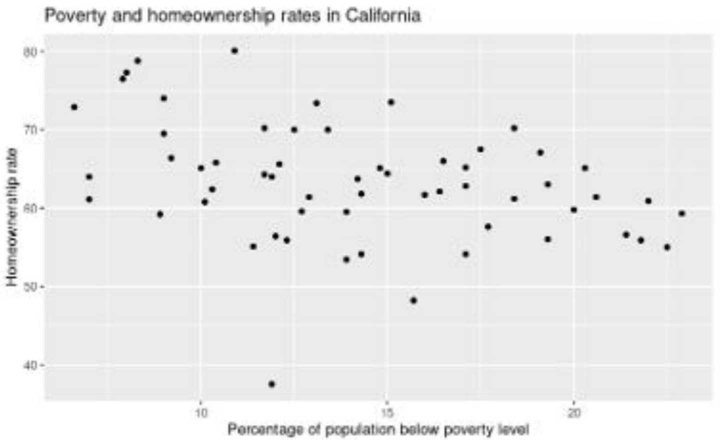
metro: Whether the county contains a metropolitan area, with levels NO, YES

poverty: Percent of population in the county below poverty level (2006-2010)

homeownership: Home ownership rate for the county (2006 – 2010)

and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
Amador County	NO	8.0	77.3



1. What are the cases in this study?

The cases in this study are the 58 counties in California.

2. Write a paragraph describing the scatterplot in context. Start with a WWWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

A study was conducted to analyze the relationship between homeownership and poverty levels in California between 2006 and 2010. The study looked into all 58 counties of California, and discovered that there is a moderate correlation between homeownership percentages and poverty levels in CA. As homeownership percentage decreases, poverty levels increase. However, since the strength of this correlation is rather weak, it's not a guarantee.

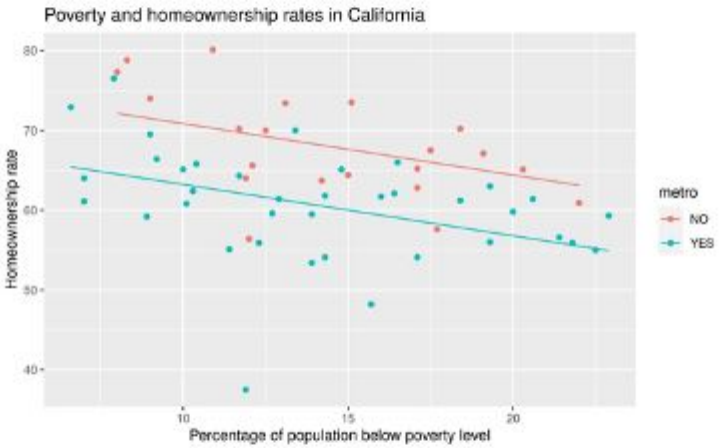
I had R create a model for predicting “homeownership” using “poverty” and “metro” as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_complete2)

Residuals:
     Min       1Q   Median       3Q      Max
-24.5043  -3.4089   0.7394   4.1270  11.9210

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  77.2778     2.9983  25.774  < 2e-16 ***
poverty      -0.6437     0.1855  -3.469  0.00102 **
metroYES     -7.6137     1.6913  -4.502 3.54e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.186 on 55 degrees of freedom
Multiple R-squared:  0.3621,    Adjusted R-squared:  0.3389
F-statistic: 15.61 on 2 and 55 DF,  p-value: 4.268e-06
```



3. Write the formula for the regression equation with a “hat” in the appropriate place.

ŷ = -0.6437P - 7.6137M + 77.2778

4. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

ŷ = -0.6437(15.7) - 7.6137(1) + 77.2778 = 59.56% homeownership rate

5. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county? Does that mean LA county's data point is above or below the line? Does that mean LA County has more homeowners than predicted or fewer?

48.2% - 59.56% = -11.36% residual in Los Angeles County. Since the data point is negative, it'll be below the regression line. This indicates that LA County has fewer homeowners than predicted.

6. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write, and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1: **The intercept, 77.2778, indicates the starting percentage of homeowners, assuming there's no poverty level or a metropolitan area in the county.**

Sentence 2: **The metropolitan area coefficient, -7.6137, is the percentage of homeownership that gets dropped when examining a county that contains a metropolitan area.**

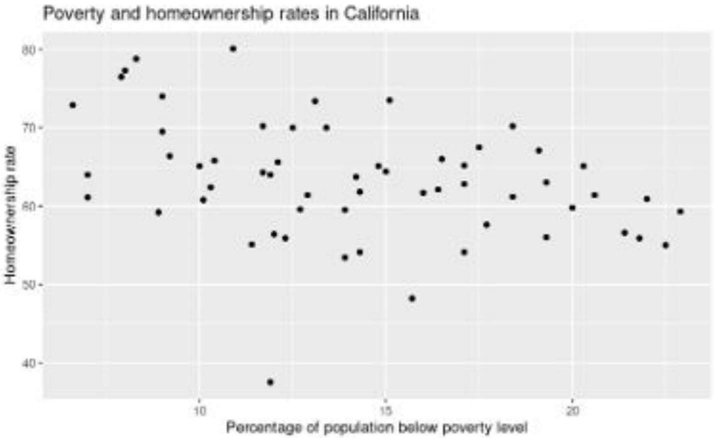
Sentence 3: **The poverty coefficient, -0.6437, is the percentage of homeownership that gets dropped depending on the poverty level percentage of the county, per each percent.**

7. Interpret the R^2 value for this model.

The R^2 value for this model is 0.3621, which indicates that 36.21% of the variance in homeownership percentages can be explained by the poverty level.

Group 9

Here is a graph of "homeownership" versus "poverty"



I have a dataset that contains information about all 58 counties in California. The data were gathered between 2006-2010. Here is a part of the code book for this dataset.

name: County name

metro: Whether the county contains a metropolitan area, with levels NO, YES

poverty: Percent of population in the county below poverty level (2006-2010)

homeownership: Home ownership rate for the county (2006 – 2010)

and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
Amador County	NO	8.0	77.3

1. What are the cases in this study?

   The cases are the counties

2. Write a paragraph describing the scatterplot in context. Start with a WWWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

This graph represents 58 counties in california inbetween 2006-2010, determining the percentages of homeownership alongside the percentage of poverty per county. Typically there is a negative, linear association between these variables: the rate of homeownership going down as the poverty percentage increases. For example, a county with a poverty percentage of 20 only has a 60% homeownership rate. Meanwhile, a county with a maybe 6% poverty rate has around a 73% homeownership rate. The relationship only only moderately strong, meaning there is some variation in the relationship between homeownership and poverty. For example, a county with a poverty percentage of 12% has a staggeringly low 38% homeownership rate.

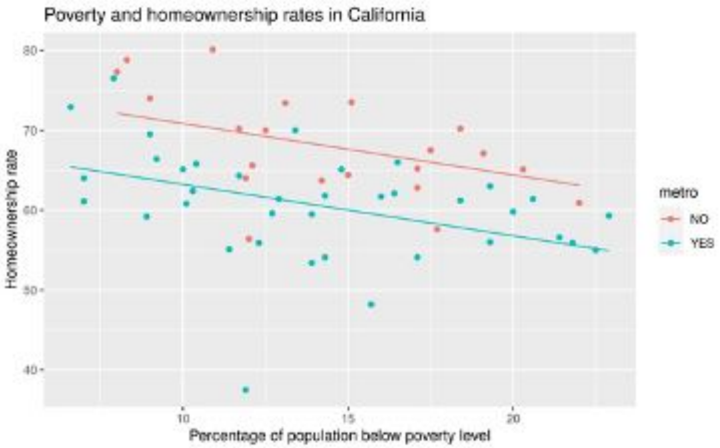
I had R create a model for predicting “homeownership” using “poverty” and “metro” as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_complete2)

Residuals:
     Min       1Q   Median       3Q      Max
-24.5043  -3.4089   0.7394   4.1270  11.9210

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  77.2778     2.9983  25.774  < 2e-16 ***
poverty      -0.6437     0.1855  -3.469  0.00102 **
metroYES     -7.6137     1.6913  -4.502 3.54e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.186 on 55 degrees of freedom
Multiple R-squared:  0.3621,    Adjusted R-squared:  0.3389
F-statistic: 15.61 on 2 and 55 DF,  p-value: 4.268e-06
```



3. Write the formula for the regression equation with a “hat” in the appropriate place.

y-hat(homeowner rate)= 77.2778-.6437(poverty)-7.6137(metroyes)

4. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

59.54001

5. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county? Does that mean LA county's data point is above or below the line? Does that mean LA County has more homeowners than predicted or fewer?

48.2-59.54001=-11.34001 This means that the LA county point is below the line, meaning that there are fewer homeowners than expected.

6. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write, and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1: If there is a poverty level of exactly 0% in any county, then the homeownership rate will automatically be 77.2778%

Sentence 2: For every 1% increase in the poverty rate, then the homeowner rates decrease by .6437 rate

Sentence 3: If the county has a metro system, then the homeownership rate will decrease by 7.6137%

7. Interpret the R^2 value for this model.

About 36% of the variability in homeownership rates is explained by the model. The remainder of the variability is determined by variables not included in the data, or by the inherent randomness of the data.

Group 10

I have a dataset that contains information about all 58 counties in California. The data were gathered between 2006-2010. Here is a part of the code book for this dataset.

name: County name

metro: Whether the county contains a metropolitan area, with levels NO, YES

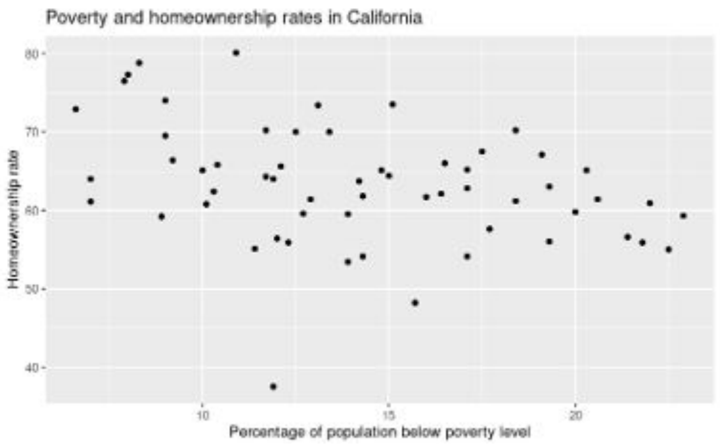
poverty: Percent of population in the county below poverty level (2006-2010)

homeownership: Home ownership rate for the county (2006 – 2010)

and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
Amador County	NO	8.0	77.3

Here is a graph of “homeownership” versus “poverty”



1. What are the cases in this study?

In this study the cases are the various different counties

2. Write a paragraph describing the scatterplot in context. Start with a WWWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

This scatterplot shows the 58 counties in california, it shows their rate of homeownership and their rate of people below the poverty line. The scatterplot doesn't demonstrate any recognizable pattern other than a very slight correlation to a somewhat inverse linear line for homeownership and percentage of people below the poverty line. For example Alameda county has 11.4% of people below the poverty line while having 55.1% of homeownership, this compared to Amador county which had a poverty line proportion of 8% while having 77% homeownership, despite the strong correlation between these two cases and the inverse linear relationship we also have cases such as alpine county which has a 13.1% poverty rate while still having a 73.4% home ownership rate.

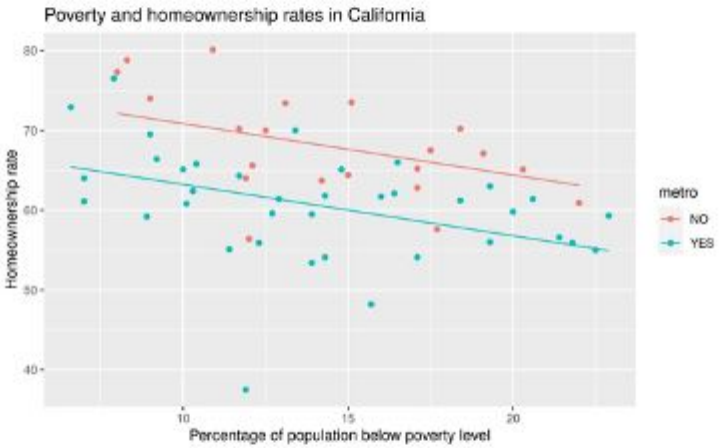
I had R create a model for predicting “homeownership” using “poverty” and “metro” as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_complete2)

Residuals:
     Min       1Q   Median       3Q      Max
-24.5043  -3.4089   0.7394   4.1270  11.9210

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  77.2778     2.9983  25.774  < 2e-16 ***
poverty      -0.6437     0.1855  -3.469  0.00102 **
metroYES     -7.6137     1.6913  -4.502 3.54e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.186 on 55 degrees of freedom
Multiple R-squared:  0.3621,    Adjusted R-squared:  0.3389
F-statistic: 15.61 on 2 and 55 DF,  p-value: 4.268e-06
```



3. Write the formula for the regression equation with a “hat” in the appropriate place.

homeownership_hat=77.2778-0.6437(poverty rate)-7.6137(yes or no; 1 or 0)

4. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

77.2778-(0.6437*0.157)-7.6137=69.5630391

5. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county? Does that mean LA county's data point is above or below the line? Does that mean LA County has more homeowners than predicted or fewer?

That means that LA county is below the expected line. LA has fewer expected homeowners that the model predicts.

6. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write, and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1: **The percentage of people below the poverty line is the number of people that fall below the poverty line in a given county divided by the total population of that county. This allows us to see how many people in a given county are living at a level where they would likely not be able to afford a house**

Sentence 2: **The percentage of homeownership is the number of people in a given county that own a home divided by the total number of people in the county. This allows us to see what percentage of a county owns a home.**

Sentence 3:

7. Interpret the R^2 value for this model.